



INTRO

OPS Roadmap

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SECTION I

Meet the Officers!

OPS Leadership



Sahil
OPS Lead Instructor



Timothy
OPS Lead Instructor



Zicong
OPS Lab Supervisor

OPS Instructors

- All your instructors are at the front!
- Everyone stand up and introduce yourself :)

SECTION II

Icebreaker

Talk to your Neighbors!

- **Name, year, major, fun fact**
- Some more ideas for conversation:
 - Where is your dream travel destination?
 - Where are you originally from?
 - Favorite cuisine
 - Least favorite school subject

SECTION III

IEEE Overview

IEEE@UCI

Who are we?

We are a student branch of the global IEEE, a professional organization with a mission to advance technology for humanity.

Where?

- Our lab is in ICS 225

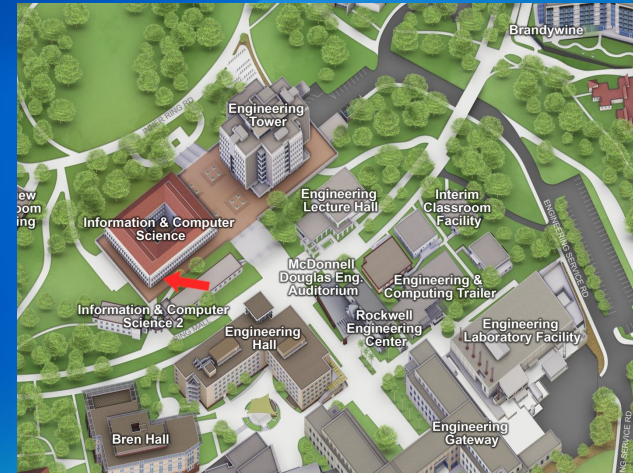
What do we offer?

- Technical Workshops
- Professional Networking
- Micromouse
- Open Project Space



The IEEE Room

- **Where?**
 - ICS 225
 - Giant Post-it "IEEE" on the window
 - You can also find us on Google Maps as "IEEE Student Branch at UCI"
- **When is it open?**
 - Check   the-lab
- **What do we have?**
 - 3D printers
 - Soldering Stations
 - Textbook Library
 - Study space
 - Lab Equipment



Click [here](#) for directions



Tentative Fall 2025 Events

Technical Workshops:

- VLSI Workshop
- IoT Workshop
- Soldering Workshop
- Keyboard Workshop

Professional Development:

- Rohde and Schwarz Info Session 10/14

Tours:

- UCI Nuclear Reactor
- UCI INRF Clean Rooms

OPS Socials:

- Hot Coco
- T-Shirt Design



SECTION IV

OPS Course Overview

Lectures and Workshops

- **Lectures**

- **Hosted every two weeks**
- Introduce topics that will be reinforced by the projects
- Are recorded, and slides are posted on the website

- **Workshops**

- Workshops are **interactive activities** to help you complete part of the project
- Only some workshops are recorded but slides are always posted
- First workshops are next week (week 4)

Lab Sessions

- Get **one-on-one support** to complete projects
- Weekly lab sessions at the IEEE Room (ICS 225)
 - Hosted by **Lab Supervisor** and **Lab Instructors**
 - **Start Week 3 (this week!)**
 - Hours posted on Canvas
- **No appointment required**



- **Weekly Announcements**

- Upcoming Projects, Lectures, and Workshops
- Important Deadlines

- **Assignment Submissions**

- Deliverables for projects are described on the assignment Canvas page as well as on the project specifications page

- **Lecture, Workshop, and Lab Schedule**

- See the entire schedule for the active Quarter



Open Project Space - 2024-2025



Syllabus

Schedule

Modules

Contact Us

Name	Email	Discord Tag
Lead Instructor		

 View Course Stream

 View Course Calendar

 View Course Notifications

To Do

-  [Welcome to Open Project Space!](#) 
Open Project Space - 2024-2025
Oct 16 at 8:32am
-  [Project 1: LED There Be Light](#) 
Open Project Space - 2024-2025
1 points |
Nov 1 at 11:59pm
-  [Project 2: 555 Piano](#) 
Open Project Space - 2024-2025
1 points |
Nov 15 at 11:59pm
-  [Project 3: RGB LED Wizard](#) 
Open Project Space - 2024-2025
1 points |
Nov 29 at 11:59pm
-  [Project 4: SunDial](#) 
Open Project Space - 2024-2025
1 points |
Dec 13 at 11:59pm

Discord

- **General Channel**

- [#ops-general](#)
- Channel where you can chat and ask general questions
- Feel free to introduce yourself here!



- **Project-Specific Questions**

- [#ops-help](#)
- Create posts and ask questions for project help here!
- Include images, code snippets, and whatever else to thoroughly explain the issue

Website

- Visit the website at ieeecs.uci.edu/ops (bookmark it now!)

Overview:

- **Course Syllabus**
 - You can use this as an overview of the full curriculum
- **Lectures and Workshops**
 - The slides and video recordings of lectures and workshops (when available) are posted on the site
- **Project Specifications**
 - Each project specification is made available at the start of the next assignment
 - The pages include links and hoverables to aid your project work

Website



Home FAQ Syllabus Projects ▾ Lectures ▾ Workshops ▾



 Apply Now



Learn. Build. Network.

We're a student-led **embedded electronics** course for **beginners**.

[Visit the Website](#)

What's in my Kit?

- **Parts**

- [Refer to the syllabus](#) for the **full list of parts** in your kit
 - Use the project specifications page to look for parts (hover your mouse over a part)
 - There may be more parts than the quantity listed in the syllabus
 - Missing a part, or a parts no longer working? Please visit the IEEE Room for a replacement

- **Nametag**

- Flip the lid of your kit, and **write your contact info** down (in case you misplace the kit)

Parts

Part Name	Qty
Jumper Wire	?
Breadboard	1
Battery, 9V	
9V Snap Connector	
Header, 2.54mm, Female, 1x2	
LED, 2V	
Slide Switch	
Resistor, 2Ω	1
Perfboard	1



OPS Student Groups

- Each one of you have been assigned a OPS Lab Instructor!
- OPS students will be divided into 11 groups (11-12 students each)

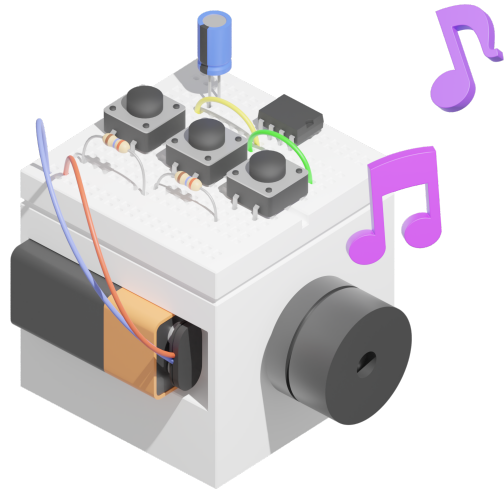
Your instructor will:

- Be your 1st point of contact & answer students' questions
- Help students in projects and the program

Student groups will be posted in threads on Discord!

Projects

- Projects are **released the same day as the lecture** (every two weeks)
- Deadlines are the night of the next lecture (11:59PM)
 - **Deadlines are recommendations** to keep pace with the course
- All projects within a quarter are due at the end of the quarter
 - (end of finals week)
- Your OPS membership is contingent upon the completion of Project 1

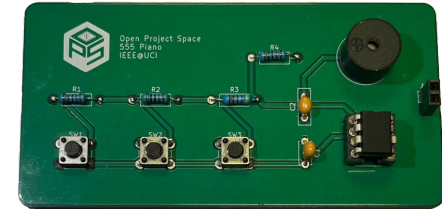


SECTION V

OPS Roadmap

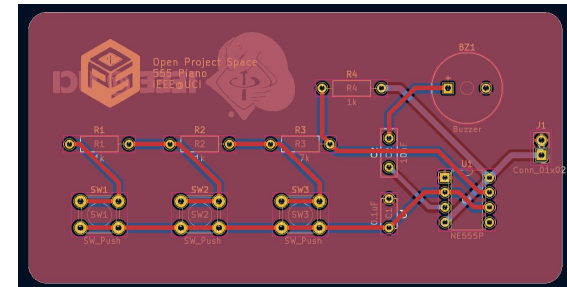
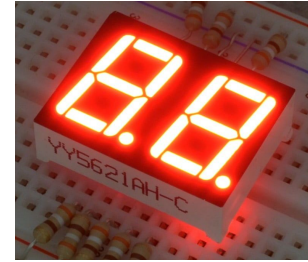
Fall Quarter Project

- **Project 1: LED there be Light**
 - Build an LED circuit with a switch and solder it to a perfboard.
- **Project 2: 555 Piano**
 - Create an electronic piano with the 555 Timer IC, and solder it to a printed circuit board.
- **Project 3: RGB LED Wizard**
 - Build and program a dimmable RGB LED using the ESP32 and potentiometers.
- **Project 4: Mini Trashcan**
 - Build a trashcan that opens automatically using an ultrasonic sensor



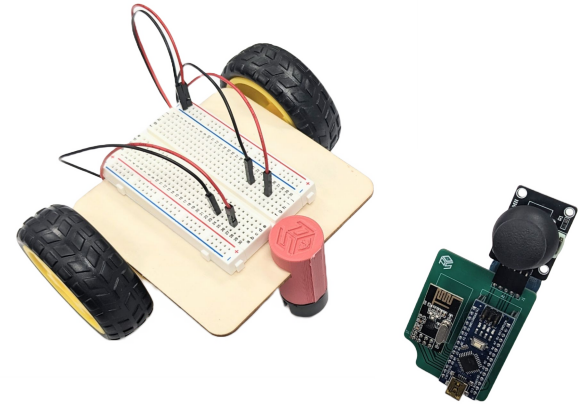
Winter Quarter Projects

- **Project 5: IPoduino**
 - Build an mp3 player and learn basic hardware communication.
- **Project 6: Weather Station**
 - Build a weather station that wirelessly transmits temperature and humidity data to an indoor display.
- **Project 7: 7-Segment Display Stopwatch**
 - Create a digital stopwatch using software interrupts and timer
- **Project 8: PCB Design With KiCad**
 - Design a PCB for the 555 Blinking LED or Piano.



Spring Quarter Projects

- **Capstone RC Rover**
 - Build and control an autonomous rover remotely with a custom PCB.
 - Any questions on any project?



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